



An effective technique for the creation of a video synopsis

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Abstract

Video synopsis involves making the content of a video brief without losing key content. The condensation ratio (CDR) is a key metric in the generation of a video synopsis. In this paper, we aim at creating a video synopsis with importance given to the CDR. Based on the CDR, many methods available in the literature have focused on the creation of a video synopsis. To generate a video synopsis with a better CDR, the proposed algorithm considers a background frame. Then, foreground information is extracted. Updating the foreground information of the initial frame is performed using the extracted foreground information of subsequent chosen frames. By adding every updated version of the initial frame, a video synopsis with an enhanced CDR is obtained. A class of methods were compared with the proposed method. Videos from standard databases, such as the UCF (University of Central Florida) Sports Action dataset, CAVIAR, and Video and Image Retrieval and Analysis Tool, were used for the evaluation of the proposed approach. The videos included highway, overpass, and sidewalk. The results demonstrated that the proposed method outperformed the other methods in terms of the CDR.

Keywords Condensation ratio · Foreground extraction · Human action · Video synopsis

1 Introduction

Video synopsis is a method to produce a concise video summary of a lengthy video. It detects and examines the objects in motion and converts long streams of video into shorter streams. The video is condensed to allow for quick retrieval and review. Video synopsis has significant applications in surveillance and video processing applications (Gayathri et al. 2020) that demand quick processing. It has emerged as an active area of research. The article is planned as follows. Section 1 offers related work in this field. Section 2 presents the material and method of the proposed approach. Then, Sect. 3 produces the results obtained with the proposed method.

Various methods proposed for video synopsis generation have been presented. The method proposed in Rav-Acha et al. (2006) applied low-level graph optimization, and the method in Pritch et al. (2008) focused on detecting moving objects and optimizing the detected objects for video condensation. The method for video condensation presented in Li et al. (2009) developed a ribbon carving (RC) model, which allows the synchronization of events and the ratio of temporal condensation. The method proposed in Pritch et al. (2009) identified outliers by observing objects with similar movements by clustering them. This method generated summaries of many objects within a short time. An online and efficient method was provided in Feng et al. (2012), which achieved high quality for endless videos with a minimum computational cost. By applying unified spatiotemporal optimization, a novel video synopsis approach was presented in Nie et al. (2013), which preserved spatiotemporal relationships and also evaded motion collisions.

By applying a video segmentation technique, a video summarization method relying on key-poses was presented in Tian et al. (2014). This method depicts the extracted objects in a video with a shape descriptor. As a whole, the method extracts the key poses by grouping the shapes. An efficient clustering method was proposed in Wu et al. (2017). This method forms clusters by combining the significant

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properties of a video. Then, the centers of all clusters were gathered as video summarization (JeanShilpa and Jawahar 2019).

A framework presented by Nanda et al. (2019) involves a three-stage person re-identification, by setting up the association between persons monitored across various camera views. This framework has combined the highlighted feature with a distance relying on correlation for re-identification inside the look-alike inlier group. An approach for global motion estimation was proposed by Wu et al. (2019) The method relied on enhanced K-means clustering algorithm, which removes the local motion vectors arising out of an unstable or shaking environment. This approach has accomplished reliable sequences by using the SURF algorithm. An object tracking framework was proposed by Dash and Patra (2019). The structure has applied a regularized robust sparse coding (RRSC) for depicting the multi-feature templates of the objects taken into account. Besides, the work has also presented a Bayesian state estimation to track the specific objects in succeeding frames.

To enhance the search speed of objects, a modified range tree method was developed in Hsia et al. (2013). This method works for videos that contain a large number of objects. To reduce the length of a video, the concept proposed in Chang and Yang (2014) provided content-based functionality. The work proposed in Zhu et al. (2014) is based on a multiple object tracking algorithm, which extracts interest blobs and tracks the objects in the video. To obtain a video synopsis, the concept presented in Huang et al. (2014) conducted a maximum a posteriori estimation using the synopsis table. This method determines the computational complexity of video synopsis generation. Relying on a content-aware framework, a high-performance video condensation system (HPVC) was provided in Zhu et al. (2015) to generate condensed videos.

A method for video synopsis was proposed in Jin et al. (2016) that quickly condenses original videos into synopsis videos with minimum collision and temporal disorder. A technique for context-aware video summarization was provided in Zhang et al. (2016), which scans all the video segments to check whether the new features with the feature correlations are sparsely represented. A technique for semantic video summarization was proposed in Mitra et al. (2017) as an entity-driven approach to video modeling. The method presented in He et al. (2017) rearranged the tubes in a video synopsis by evaluating the collision relationships between them. An approach for video synopsis was proposed in Li et al. (2016). By applying seam carving video tubes, the method minimized the redundancy of object motion. An effective method for retrieving videos (Thomas et al. 2017) provided a synopsis for large video databases. The technique proposed in Bhaumik et al. (2016) demonstrated the quality of the video summary by eliminating duplicate frames.

2 Materials and methods

The present study aims at obtaining a video synopsis for a set of videos taken from standard databases and real-time videos. By considering a set of relevant frames in the video, the approach represents the human actions in the video by condensing the content. The condensation ratio (CDR) is used as a metric to evaluate the proposed method. Figure 1 gives the block diagram of the proposed approach.

The algorithms used in the proposed method are as follows: Algorithm 2.1 provides the main routine, which calls various routines for the generation of the video synopsis. To extract the foreground information from the current frame, Algorithms 2.2 and 2.3 are invoked. Algorithm 2.2 calculates the threshold for the foreground information. Algorithm 2.3 retrieves the position of the foreground in the current frame by calling Algorithm 2.4. Then, Algorithm 2.5 updates the foreground information of the initial frame with the foreground information of the current frame.

A frame is considered initially as a background frame. A set of frames are chosen depending on the number of output frames required to generate the video synopsis. The foreground information for the chosen set of frames is extracted. The position information of the current foreground is obtained by performing a morphological operation and applying a threshold. This position information is used to extract the foreground information, which includes color in the frame. Then, the foreground information of the initial frame is updated each time with the extracted foreground information of the frame. The video synopsis is generated by adding each of the updated versions of the initial frame.

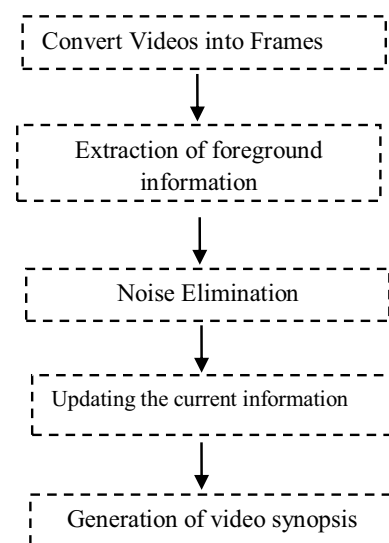


Fig. 1 Block diagram of the proposed approach

2.1 Algorithm for video synopsis generation

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Step1:   Get input video information (k)
Step2:   frame = initial frame
Step3:   for i = ifr to nth frame
Step4:     read current frame (k(i).name) to color_frame;
Step5:     convert color information to gray information (imc) of the current frame
Step6:     extract foreground information by applying Steps 9 to 11
Step7:     set the structural element SE;
Step8:     perform a morphological operation and apply threshold (BW1);
Step9:     currentforeground = calframe(im,BW1);
Step10:    frame=frame_update(currentforeground,frame,color_frame,frsize);
Step11:    i=i+ incfr //incfr is next frame number
Step12:  end
    
```

2.2 Algorithm for threshold (d)

Step 1: $thrs = \frac{1}{rr+rc} \left(\sum_{rc} \max(d) + \sum_{rr} \max(d^T) \right)$

Step 2: $d_f = \{ \delta(i,j) | 1 \leq i \leq rr, 1 \leq j \leq rc \}$, where

$$\delta(i,j) = \begin{cases} 0, & \forall d(i,j) \in d : d(i,j) \leq thrs \\ 1, & otherwise \end{cases}$$

2.3 Algorithm for calframe (im1, BW1)

Step 1: $[findrow \ findcol] = \text{find_test}(BW1, 1)$

Step 2:

$$currentimval(row, col) = \begin{cases} im1(row, col), & \exists row = findrow(i), col = findcol(i), 1 \leq i \leq size(findrow) \\ 0, & otherwise \end{cases}$$

2.4 Algorithm for find_test (imf, fval)

Step 1: $[imfrow, imfcol] = \text{size}(imf)$

Step 2: $fr = 1$;

Step 3: $findrow(fr) = \{i, \forall imf(i,j) = fval | 1 \leq i \leq imfrow, 1 \leq j \leq imfcol\}$

Step 4: $fr = 1$;

Step 5: $findcol(fr) = \{j, \forall imf(i,j) = fval | 1 \leq i \leq imfrow, 1 \leq j \leq imfcol\}$

2.5 Algorithm for frame_update (currentim, frame, color_frame, frsize)

Step 1: $updatedframe = frame$

Step 2: $updatedframe(i,j,k) = \{colorframe(i,j,k), \forall currentim(i,j) \neq 0, 1 \leq i \leq frsize(1), 1 \leq j \leq frsize(2), 1 \leq k \leq 3\}$

$updatedframe(i,j,k) = \{colorframe(i,j,k), \forall currentim(i,j) \neq 0, 1 \leq i \leq frsize(1), 1 \leq j \leq frsize(2), 1 \leq k \leq 3\}$

Fig. 2 Sample input frames obtained from real-time video 1



Fig. 3 Condensed frames obtained from real-time video 1 of Fig. 2

Fig. 4 Sample input frames obtained from real-time video 2

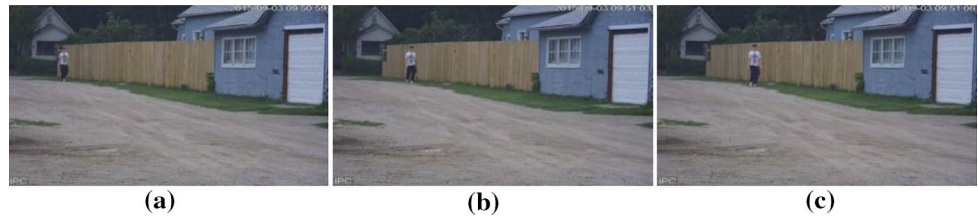


Fig. 5 Condensed frames obtained from real-time video 2 of Fig. 4



Fig. 6 One of the condensed frames obtained from VIRAT and RC (Li et al. 2009)



Fig. 7 One of the condensed frames obtained from CAVIAR and UCF sports action dataset

3 Results and discussion

Videos with distinct foreground and background from standard databases [real-time videos, CAVIAR, UCF sports action dataset, video and image retrieval and analysis tool (VIRAT), and RC (Li et al. 2009)] were considered for the evaluation of the proposed approach. Figure 2 shows a sample input frame selected from real-time video 1. The resulting condensed frames obtained for the videos selected

Table 1 Calculation of ratio T_R for various scenarios

Scenarios	Input video time in secs (T_{IV})	Output video time in secs (T_{OV})	Ratio (T_R)
Real-time video 1	18	1	0.056
Real-time video 2	31	2	0.065
VIRAT	64	3	0.047
UCF sports action dataset	12	1	0.083

Table 2 Computation of ratio S_R for various scenarios

Scenarios	Input video size in MB (S_{IV})	Output video size in MB (S_{OV})	Ratio (S_R)
Real-time video 1	2.72	0.448	0.165
Real-time video 2	17.7	1.23	0.069
VIRAT	37.1	1.98	0.053
UCF sports action dataset	1.24	0.208	0.168

from real-time video 1 are shown in Fig. 3. Figure 4 shows a sample input frame selected from real-time video 2. The resulting condensed frames obtained for the videos selected from real-time video 2 are shown in Fig. 5. The condensed frames for the videos selected from VIRAT and RC (Li et al. 2009) are shown in Fig. 6. The condensed frames for the videos selected from CAVIAR and the UCF Sports Action dataset are shown in Fig. 7.

The ratio T_R of the output video time duration T_{OV} and input video time duration T_{IV} is calculated as,

$$T_R = \frac{T_{OV}}{T_{IV}} \quad (1)$$

The ratio T_R in Eq. (1) obtained by the proposed method is shown in Table 1.

The ratio S_R of the output video size S_{OV} and input video size S_{IV} is computed as,

$$S_R = \frac{S_{OV}}{S_{IV}} \quad (2)$$

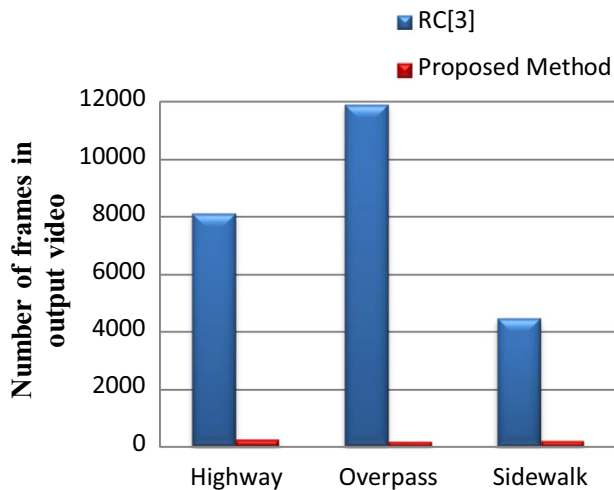
The ratio S_R in Eq. (2) obtained by the proposed method is presented in Table 2.

Table 3 Evaluation of proposed method with different videos used in RC (Li et al. 2009)

Video	No of frames in the input video		No of frames in the output video		Condensation ratio (CDR)	
	RC (Li et al. 2009)	Proposed method	RC (Li et al. 2009)	Proposed method	RC (Li et al. 2009)	Proposed method
Highway	23,950	23,950	8114	240	2.95	99.79
Overpass	23,950	23,950	11,880	175	2.02	136.85
Sidewalk	7950	7950	4462	190	1.78	126.05

Table 4 Evaluation of proposed method with various methods for overpass video with 23,950 frames

Method	No of frames in output video	Condensation ratio (CDR)
RC (Li et al. 2009)	11,880	2.02
ST (Feng et al. 2012)	201	119
HPVC (Zhu et al. 2015)	1004	23.85
Proposed method	175	136.85

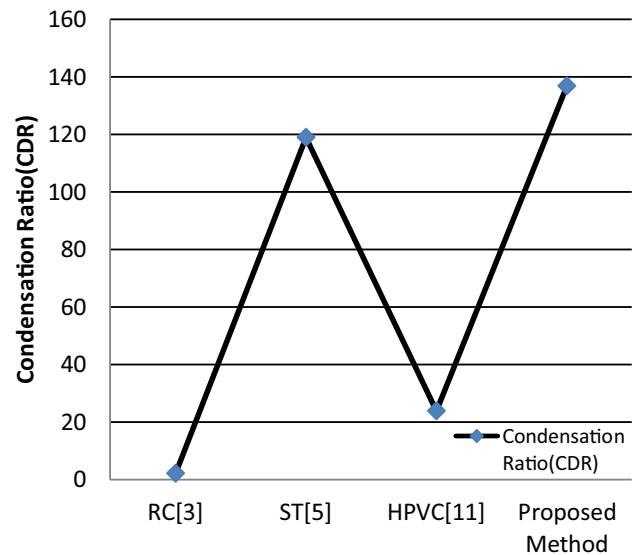
**Fig. 8** Comparison of proposed method with different videos used in RC (Li et al. 2009)

Results shown in Tables 1 and 2 illustrate that the proposed method has generated a video synopsis with minimum time and size when compared with the input videos.

The proposed method was evaluated with regard to the CDR, which is computed as

$$CDR = \frac{\text{Number of frames in the input video}}{\text{Number of frames in the output video}} \quad (3)$$

With regard to the CDR calculated using Eq. (3), the proposed method was compared with method RC (Li et al. 2009) for highway, overpass, and sidewalk videos. The

**Fig. 9** Comparison of the proposed method with other methods for an overpass video with 23,950 frames

results are shown in Table 3 clearly indicate that the proposed method achieved a better CDR than method RC (Li et al. 2009).

Based on the CDR, the proposed method was compared with the methods RC (Li et al. 2009), ST (Feng et al. 2012), and HPVC (Zhu et al. 2015) for the overpass video. Table 4 shows the results of the evaluation, which illustrates that the proposed method attained a better CDR than the other methods.

For various videos, such as highway, overpass, and sidewalk, the proposed method was compared with the method RC (Li et al. 2009). The results of the comparison are shown graphically in Fig. 8. In the graph, different videos are plotted on the x-axis and the number of frames in the output video are plotted on the y-axis. The results demonstrate that the proposed method generated the video synopsis with the minimum number of frames compared with RC (Li et al. 2009).

An overpass video containing 23,950 frames was selected to compare the proposed method with the methods RC (Li et al. 2009), ST (Feng et al. 2012), and HPVC (Zhu et al.

2015). A graphical representation of the results obtained is shown in Fig. 9. In this graph, various methods are plotted on the x -axis and the CDR is plotted on the y -axis. Based on the results, the performance of the proposed method is exemplary for various categories of videos on standard databases.

4 Conclusion

The proposed method produced a video synopsis with a better CDR. This research was conducted based on its importance and applications. The method started by choosing the foreground information of the initial frame and ended by adding every updated version of the initial frame. Thus, a video synopsis with an improved CDR was obtained. The proposed method was compared with a similar set of methods. The results clearly demonstrated that the proposed method was better than the other methods with regard to the CDR. The proposed work can be protracted to videos that contain scenes with shadows, occlusions and camera movements.

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